MIL-C-915/48A
1 August 1972
SUPERSEDING[1/]
MIL-C-915/48(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 2SWF

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper conductor, ASTM B-286 size 22-7, tin coated. Maximum diameter 0.033 inch.

(A) Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Nominal thickness 0.010 inch. Colored insulation, one black and one white conductor for each pair.

(A) Third - Clear polyamide jacket on each conductor, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.

Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form each pair.

Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.

Sixth - Braided shield of AWG No. 34 or 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.

(A) Seventh - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness of 0.003 inch. Standard Identification Code, applied by Method 2.

- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code, applied by Method 2, on inner tape.

(A) Eighth - The required number of pairs (see details) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair No. 1, from center outward. Non-fibrous fillers shall be employed.

Ninth - Synthetic rubber compound filled binder tape applied helically with overlap. (Surface of tape shall be such that cable jacket will adhere to it).

(A) Tenth - Arctic type polychloroprene jacket (black). (See details for thickness.) Cable surface marking required.

Details

Type and size	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter	
			Minimum (inch)	Maximum (inch)
2SWF-3	3	0.100	0.600	0.625
2SWF-4	4	.090	.600	.625
2SWF-7	7	.110	.780	.815

[1/] Requirements formerly covered on MIL-C-24145/8.

(A) Denotes changes

FSC 6145

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(A) Conductor resistance - Ohms/1000 ft. at 25°C., maximum	17.71
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Group A:	
Visual and dimensional	
(A) Hydrostatic (open end) - Leakage at 1000 psi, cu. in., 6 hrs., (test each shipping length)	Zero
Capacitance	
Mutual capacitance - At 1 mHz, pF/ft., maximum	30
Capacitance unbalance - Percent, maximum	8
Characteristic impedance - At 1 mHz, ohms	75 ± 5
Group B:	
Attenuation - At 3 mHz, db/100 ft., maximum	3
(A) Cold working (minus 54° ± 2° C.)	(No damage)
(A) Drip - At 75° ± 1°C.,	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	60
Permanence of printing (jacket) - Cycles, minimum	250
Cable filler removability	

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical plus all of Groups A, B and C, plus the following:

(A) Cable aging and compatibility (95° ± 3°C.)	No failure
(A) Cold working (minus 54° ± 2°C.)	No failure

UNIT ORDERING LENGTHS: All sizes: 1000 feet (nominal)

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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